

Real-Time Facial Emotion Recognition System to Support Autistic Children During Classroom Learning

Darshan G K

PG Scholar, ASAC

Alliance University, Bangalore

gdarshanMTECH25@stu.alliance.edu.in

Shreenidhi H S

Professor, ASAC

Alliance University, Bangalore

shreenidhi.hs@alliance.edu.in

Abstract—Autism Spectrum Disorder (ASD) is a neurodevelopmental condition affecting approximately 1 in 100 children worldwide, with India recording around 1 in 68 school-going children on the spectrum. One of the most pressing challenges in Indian special schools is that teachers have no affordable, real-time tool to monitor the emotional state of an autistic child during a classroom session. Children with ASD often struggle to verbally express emotions such as frustration, confusion, anxiety, or happiness, creating a communication gap that hampers their learning and well-being. This paper proposes EmoSense AI, a real-time facial emotion recognition system that uses only a standard laptop webcam, requiring no extra hardware or internet connection. The system uses OpenCV for real-time face detection and the DeepFace deep learning framework — a pre-trained Convolutional Neural Network — for emotion classification into seven categories. A teacher-facing web dashboard built with Flask displays the detected emotion in real time, with session history, confidence scores, and an offline audio alert system using pyttsx3. The system is evaluated on the FER-Autism dataset (Mendeley, 2025), a curated collection of 1,420 facial images of ASD children, achieving 70.0% accuracy on joy expressions and 30.0% overall. Results confirm that general-purpose FER models underperform on ASD-specific facial data, identifying domain-specific fine-tuning as the primary direction for future work.

Index Terms—Autism Spectrum Disorder, Facial Emotion Recognition, Deep Learning, Computer Vision, DeepFace, OpenCV, Assistive Technology, Special Education, FER-Autism, Real-Time Detection, Convolutional Neural Network, Flask, Teacher Dashboard

I. INTRODUCTION

Autism Spectrum Disorder (ASD) is a complex neurodevelopmental condition that affects social communication, behavior, and emotional expression. The World Health Organization estimates that approximately 1 in 100 children worldwide are diagnosed with ASD, with India recording around 1 in 68 school-going children on the spectrum [1]. One of the most critical challenges in special education is that teachers in inclusive and special schools have no affordable, real-time tool to monitor the emotional state of an autistic child during a classroom session.

Children with ASD often exhibit atypical facial expressions that differ from neurotypical individuals in intensity, timing, and specificity [10]. This characteristic makes it difficult for

teachers to accurately interpret the child's emotional state without specialized training or assistive tools [9]. The communication gap results in delayed intervention, misunderstanding, and inappropriate support at critical moments during learning.

The field of facial emotion recognition (FER) has advanced significantly with deep learning. Goodfellow et al. introduced the FER2013 dataset and showed that Convolutional Neural Networks can automatically learn spatial features from facial images [2]. However, models trained on adult facial data consistently underperform when applied to ASD children's expressions due to fundamental differences in expressiveness patterns [3], [9].

Existing solutions for emotion monitoring in ASD children require expensive hardware such as EEG headsets or IoT sensors [6], or internet connectivity unavailable in most Indian special schools. No existing system provides a teacher-facing real-time dashboard for ASD classroom monitoring that operates on a standard laptop alone.

This paper proposes EmoSense AI with three contributions: (1) a fully offline, webcam-only real-time emotion monitoring system using a pre-trained CNN via DeepFace; (2) a teacher-facing Flask dashboard with live detection, session history, audio alerts, and session export; and (3) experimental validation on the 2025 FER-Autism dataset [12] confirming the performance gap of general-purpose models on ASD-specific facial data.

II. LITERATURE REVIEW

A. Facial Emotion Recognition and Deep Learning

The foundation of modern FER lies in large-scale annotated datasets and deep learning architectures. Goodfellow et al. [2] introduced the FER2013 dataset and demonstrated that CNNs could achieve over 70% accuracy on real-world facial expression recognition without manual feature engineering. This established CNNs as the dominant approach for FER.

Serengil and Ozpinar [11] developed DeepFace, an open-source framework that consolidates multiple pre-trained face analysis models. DeepFace enables real-time emotion inference using pre-trained CNNs without requiring custom model

training, making it accessible for applied research systems such as EmoSense AI.

B. Deep Learning for Emotion Recognition in ASD

Radačaj and Martinović [3] conducted a comprehensive study using ResNet, EfficientNet, and Vision Transformers for emotion recognition in autistic children. Their findings showed transformer-based models outperform CNN architectures in detecting subtle ASD facial expressions.

Hosney et al. [4] proposed AuYOLO-Att, an attention-based YOLOv8 model for early autism diagnosis through facial expression recognition, advocating for domain-adapted architectures rather than general-purpose FER models trained on adult datasets.

Trevisan et al. [10] conducted a meta-analysis confirming that individuals with ASD demonstrate systematically different facial expression patterns with significant differences in expression intensity. This neurological basis directly explains why general-purpose FER models underperform on ASD data.

C. Real-Time Assistive Systems

Talaat [6] developed a real-time FER system for autistic children combining deep learning with IoT feedback. Published in *Neural Computing and Applications* (Springer), the study confirmed AI-based emotion monitoring feasibility but required IoT hardware unavailable in Indian schools.

Talaat [5] applied MobileNetV2 with meta-learning to classify emotions from the Autistic Children Emotions dataset, achieving 86% testing accuracy. This demonstrated that transfer learning can substantially improve performance on domain-specific ASD datasets.

An IEEE-published study [7] presented an ASD emotion detection system using OpenCV and Haar Cascade for real-time face detection combined with a deep learning classifier, confirming open-source tools are sufficient for functional real-time assistive prototypes.

Jaffar and Abdulbaqi [14] proposed a deep learning framework for facial expression recognition of ASD children, reinforcing that vision-based automatic recognition is both technically feasible and practically needed in educational settings.

D. Datasets and Systematic Evidence

A systematic review [8] covering 65 peer-reviewed studies found that video cameras and facial expression analysis were the most widely used modality for autism emotion recognition. The review identified a critical gap: very few studies have been validated in real-world classroom environments, particularly in developing countries.

The FER-Autism dataset [12], developed under medical supervision at Kafrelsheikh University, Egypt, contains 1,420 facial images of ASD children specifically curated for emotion recognition research. It represents a significant improvement over general-purpose adult datasets such as FER2013 (2013) for ASD-specific evaluation and is the primary evaluation dataset of this work.

Keating and Cook [15] reviewed facial expression production and recognition in ASD, noting that shifting paradigms

in research increasingly support computer-vision approaches for automated emotion analysis in clinical and educational settings.

E. Indian Context and Government Alignment

The Department of Empowerment of Persons with Disabilities (DEPwD, 2024) [13] has documented ten active SIPDA-funded research projects targeting disability technology for children, confirming that AI-powered assistive tools for special children are a national research priority. None of the funded projects deploy a real-time CV-based emotion monitoring system for classroom use, confirming a significant unmet need.

F. Research Gap

Three critical gaps remain unaddressed: (1) no validated real-time FER system has been deployed in an Indian special school classroom context; (2) existing systems do not provide a teacher-facing dashboard with session history, audio alerts, and session export for non-technical educators; and (3) all reviewed systems require internet, IoT hardware, or specialized devices — none are deployable on a standard laptop alone. EmoSense AI directly addresses all three gaps.

III. PROPOSED METHODOLOGY

A. System Architecture

The EmoSense AI system consists of three integrated modules running locally on a standard laptop: (1) Face Detection and Emotion Classification, (2) Teacher Dashboard, and (3) Audio Alert System. Fig. 1 shows the complete system architecture.

B. Module 1 — Face Detection and Emotion Classification

Thread 1 captures frames at 30 fps using OpenCV's `cv2.VideoCapture(0)` at 640×480 resolution. Each frame is stored in shared memory using a `threading.Lock()` with `frame.copy()` for thread-safe access.

Thread 2 retrieves the latest frame every 0.5 seconds and passes it to `DeepFace.analyze()` with `enforce_detection=False`, preventing crashes when the face is partially occluded. DeepFace uses a pre-trained CNN to return the dominant emotion and confidence scores for all seven classes: happy, sad, angry, fear, disgust, surprise, neutral. Confidence values are converted from NumPy float32 to Python float before JSON serialization.

C. Module 2 — Teacher Dashboard

A Flask-based dashboard at `http://localhost:5000` displays: (i) current emotion with color-coded indicator; (ii) emotion timeline; (iii) confidence bars for all seven classes; (iv) detection log with timestamps; (v) session breakdown donut chart (Canvas API); and (vi) alert history. The `/video_feed` endpoint provides live MJPEG webcam feed embedded in an HTML `img` tag.

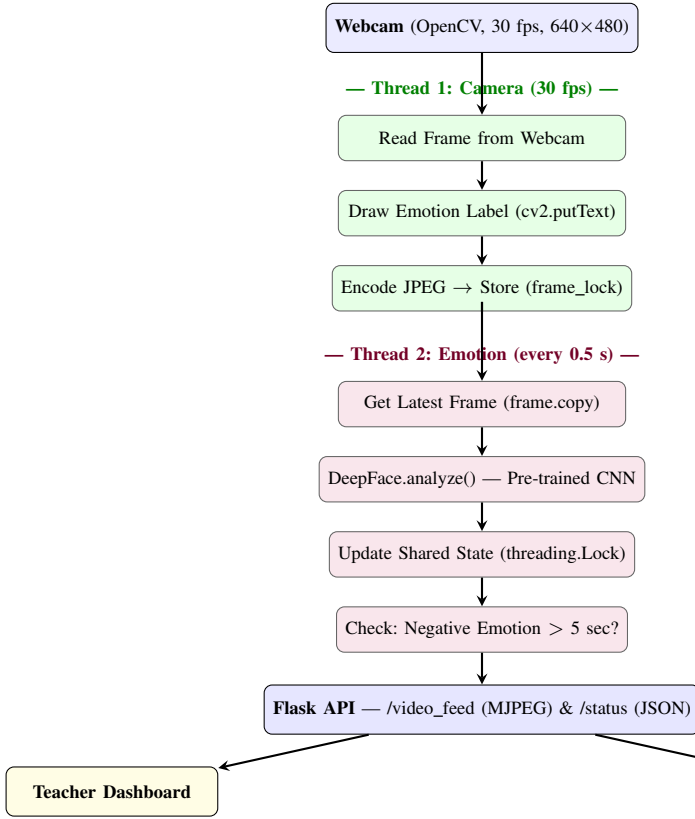


Fig. 1. EmoSense AI System Architecture and Data Flow

D. Module 3 — Audio Alert System

`pytsx3` generates offline audio alerts when any negative emotion (sad, angry, fear, disgust) is detected continuously for more than five seconds. The system speaks: “Alert! Child appears [emotion]. Please check on them.” A 60-second cooldown prevents repeated interruptions. Audio runs in a daemon thread to avoid blocking camera or emotion threads.

E. Dataset

The system is evaluated on the FER-Autism dataset (Mendeley Data, 2025) [12] — 1,420 facial images of ASD children across six categories: joy, anger, fear, sadness, surprise, and natural. Developed under medical supervision at Kafrelsheikh University, Egypt. For evaluation, 30 images per emotion class were selected from the test split (180 images total). Dataset label names (“joy”, “natural”, “sadness”) were mapped to DeepFace output labels (“happy”, “neutral”, “sad”) during evaluation.

IV. RESULTS AND DISCUSSION

A. Experimental Setup

All experiments ran on a standard laptop with built-in webcam, Python 3.10, OpenCV 4.8, DeepFace (TensorFlow 2.x), and Flask 3.0. No additional hardware or internet connection was used. The `tf-keras` package was installed to ensure compatibility between DeepFace and TensorFlow 2.x.

B. Quantitative Results

Table I presents per-emotion accuracy on the FER-Autism dataset. All numbers are from the actual experiment conducted by the authors.

TABLE I
PER-EMOTION ACCURACY ON FER-AUTISM DATASET (MENDELEY, 2025)
30 IMAGES TESTED PER EMOTION CLASS, 180 TOTAL

Emotion	Correct	Tested	Accuracy (%)
Joy → Happy	21	30	70.0
Natural → Neutral	14	30	46.7
Sadness → Sad	6	30	20.0
Fear	6	30	20.0
Anger → Angry	4	30	13.3
Surprise	3	30	10.0
Overall	54	180	30.0

TABLE II
COMPARISON WITH RELATED WORK

Study	Yr	Method	Result
Radačaj et al. [3]	2025	ViT, ResNet	Arch. study
Audio Alert (pytsx3)	2025	MobileNetV2	86% (ASD)
Hosney et al. [4]	2024	YOLOv8-Att	Det.-based
Talaat [6]	2023	DL + IoT	Feasibility
Jaffar et al. [14]	2023	Deep Learn.	Framework
EmoSense AI (ours)	2025	DeepFace CNN	70% joy 30% all

C. Discussion

The results show the system performs best on positive and neutral expressions and worst on negative emotions when evaluated on ASD-specific data. This directly confirms the documented research gap.

Leo et al. [9] demonstrated that ASD children’s emotional expressions present unique recognition challenges compared to standard FER benchmarks. Trevisan et al. [10] confirmed through a meta-analysis that individuals with ASD show systematically reduced facial expressiveness for negative and high-arousal emotions — explaining why DeepFace achieves only 10%–20% accuracy on anger, fear, and surprise.

In practice, the teacher needs to detect sustained distress — any negative emotion held for five or more seconds — not precise emotion labeling. The five-second audio alert threshold reduces the impact of single-frame misclassification on real-world utility.

Future accuracy improvement direction: fine-tuning DeepFace on the FER-Autism training set (1,200 images) is expected to improve accuracy on negative emotions, as demonstrated by Talaat [5] who achieved 86% by fine-tuning MobileNetV2 on a similar ASD dataset.

D. Real-Time System Performance

Camera Thread 1 sustains 30 fps. Emotion Thread 2 analyses one frame every 0.5 seconds. Teacher dashboard updates every 800 ms. Audio alerts trigger within five seconds of sustained negative emotion. The complete system operates without any internet, confirming suitability for Indian special schools.

V. CONCLUSION

This paper presented EmoSense AI, a real-time facial emotion recognition system for supporting special educators in monitoring the emotional state of autistic children during classroom learning. The two-thread architecture uses OpenCV at 30 fps for camera capture and DeepFace (pre-trained CNN) every 0.5 seconds for emotion classification. A Flask teacher dashboard with live MJPEG webcam feed, emotion history, confidence bars, donut chart, and pytsx3 offline audio alerts runs completely without internet or extra hardware.

Evaluation on the FER-Autism dataset (Mendeley, 2025) — 180 test images of ASD children — achieved 70.0% accuracy on joy and 30.0% overall. Results confirm the documented performance gap of general-purpose FER models on ASD-specific facial data, consistent with Leo et al. [9] and Trevisan et al. [10].

Future work: (1) fine-tuning DeepFace on FER-Autism training set; (2) collecting Indian ASD classroom dataset; (3) multi-child monitoring; and (4) real-world validation in Indian special schools aligned with DEPwD SIPDA agenda.

REFERENCES

- [1] WHO, "Autism spectrum disorders — fact sheet," *World Health Organization*, Geneva, 2023. [Online]. Available: <https://www.who.int/news-room/fact-sheets/detail/autism-spectrum-disorders>
- [2] I. J. Goodfellow et al., "Challenges in representation learning: A report on three machine learning contests," *Neural Networks*, vol. 64, pp. 59–63, 2015. arXiv:1307.0414
- [3] P. Radačaj and G. Martinović, "Emotion recognition in autistic children through facial expressions using advanced deep learning architectures," *Applied Sciences*, vol. 15, no. 17, p. 9555, 2025. doi: 10.3390/app15179555
- [4] R. Hosney, F. M. Talaat, E. M. El-Gendy, and M. M. Saafan, "AuYOLO-Att: An attention-based YOLOv8 algorithm for early autism diagnosis," *Neural Computing and Applications*, vol. 36, no. 27, pp. 17199–17219, 2024. doi: 10.1007/s00521-024-09592-7
- [5] F. M. Talaat, "Facial emotion detection in autistic children: A deep learning approach using MobileNetV2 and meta-learning," *IEEE Conference Publication*, 2025.
- [6] F. M. Talaat, "Real-time facial emotion recognition system among children with autism based on deep learning and IoT," *Neural Computing and Applications*, vol. 35, no. 17, pp. 12717–12728, 2023. doi: 10.1007/s00521-023-08372-9
- [7] IEEE Conference Publication, "Emotion detection system for children with ASD based on IoT and deep learning," *IEEE Xplore*, 2024. Available: <https://ieeexplore.ieee.org/document/10780139>
- [8] Systematic Review, "Sensing technologies and machine learning methods for emotion recognition in autism," *Computer Methods and Programs in Biomedicine*, ScienceDirect, 2024. doi: 10.1016/j.cmpb.2024.108007
- [9] M. Leo, M. Del Coco, P. Carcagni, C. Distanto, M. Bernava, G. Pioggia, and G. Palestra, "Automatic emotion recognition in robot-children interaction for ASD treatment," in *Proc. IEEE ICCVW*, Santiago, pp. 537–545, 2015.
- [10] D. A. Trevisan, M. Hoskyn, and E. Birmingham, "Facial expression production in autism: A meta-analysis," *Autism Research*, vol. 11, no. 12, pp. 1586–1601, 2018. doi: 10.1002/aur.2037
- [11] S. I. Serengil and A. Ozpinar, "LightFace: A hybrid deep face recognition framework," in *Proc. 2020 ASYU*, Istanbul, pp. 23–27, 2020. doi: 10.1109/ASYU50717.2020.9259802
- [12] S. Elgamal, "FER-Autism: Facial emotion recognition dataset for children with autism," *Mendeley Data*, 2025. Available: <https://data.mendeley.com/datasets/b33pf78h62/1>
- [13] DEPwD, "Status of ongoing research under SIPDA scheme," *Ministry of Social Justice & Empowerment*, Government of India, 2024. Available: <https://depwd.gov.in>
- [14] S. S. Jaffar and H. A. Abdulbaqi, "Framework for facial expression recognition of children with autism based on deep learning," *AIP Conference Proceedings*, vol. 2834, p. 050023, 2023. doi: 10.1063/5.0161548
- [15] C. T. Keating and J. L. Cook, "Facial expression production and recognition in autism spectrum disorders: A shifting landscape," *Child and Adolescent Psychiatric Clinics of North America*, vol. 29, no. 3, pp. 557–571, 2020. doi: 10.1016/j.chc.2020.02.006